SCMS School of Engineering and Technology

Department of Computer Science and Engineering Project Planning & Proposal Form

CSD 415 PROJECT PHASE I -BATCH 2023-2027

Project Title: **SecureHealth Chain**: A Lightweight Blockchain-Based Federated Learning Platform for Secure Health Records

Group No & Members:

GROUP 4

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Class: S7 B.Tech CS1

Date:20-07-2026

1. Purpose of project

Electronic Health Records are increasingly digitized, but centralized storage models expose patient data to security breaches, unauthorized access, and single points of failure, while blockchain-federated learning frameworks such as EPP-BCFL secure records using techniques that are computationally heavy and unnecessary for students and small-scale prototypes. As a result, implementing full frameworks within a limited final-year timeline can be resource-intensive and impractical.

The purpose of this project is to develop SecureHealth Chain, a lightweight blockchain-based federated learning platform that enables hospitals to train disease-prediction models and manage Electronic Health Records through role-based portals. The platform focuses on essential functionality such as federated model training, blockchain hash storage, AI-based risk prediction, and QR-based record verification while maintaining minimal computational overhead.

2. Project Outline

Module 1 – User Authentication & Role Management

- Secure registration and login for Admin, Doctor, and Patient roles.
- Role-based dashboard access and permissions.
- Patient EHR upload and profile management.
- Doctor interface for reviewing patient records and predictions.

Module 2 – Local AI Model & Federated Learning Client

- Local training of disease-prediction models on hospital/node-level EHR data.
- Federated Learning client setup using Flower/TensorFlow Federated.
- Model update generation without sharing raw patient data.
- Communication of encrypted model updates to the FL aggregation server.
- Local model evaluation and accuracy reporting before submission.

Module 3 – Federated Aggregation & Blockchain Hash Storage

- Central FL server aggregating model updates (Federated Averaging).
- Hashing of model updates and medical records.
- Storage of hashes on a Ganache-based Ethereum blockchain for tamper-evident verification.
- Global model update and redistribution to nodes.
- Verification of stored hashes against incoming updates before acceptance.

Module 4 – AI Health Risk Dashboard & QR Verification

- AI Health Risk Prediction Dashboard showing disease risk, model accuracy, and blockchain verification status.
- QR code generation embedding the blockchain transaction hash for each medical record.
- QR-based scan-and-verify workflow for authorized record integrity checks.
- Last-updated timestamps and verification logs.

3. Project outcomes/key deliverables at the end of VIII Semester

- Journal Publication ☑
- Conference Publication
- Patent Publication
- APP / Web Hosting ☑
- Hardware Product
- Project Funding
- Others (specify details)

4. Resource Allocation

4.1 Proposed budget

Cloud/VPS Hosting (Testing & Deployment)	2,500
Documentation & Printing	1,000
Miscellaneous Expenses	1,500

Total Estimated Cost ₹5000

4.2 Technology Used

Software

- React.js
- Flask
- MySQL
- Flower (Federated Learning Framework)
- Ethereum Ganache
- Solidity
- TensorFlow / Scikit-learn
- QR Code Library

Programming Languages

- Python
- JavaScript
- SQL
- Solidity
- HTML
- CSS

4.3 Other resources (Kit, Sensors, Cloud subscription, etc)

Infrastructure

- Local Development System
- Ganache Ethereum Blockchain
- MySQL Database Server

Cloud Resources

- VPS Hosting (Optional)
- GitHub Repository
- Domain Name (Optional)

Development Tools

- Visual Studio Code
- Git & GitHub
- Docker Desktop (Optional)
- Postman
- MetaMask Wallet

5. Timeline with Milestones (August 2023 to May 2024) Ganttcharts

ACTIVITIES	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR
Literature Survey	✓	✓							
Requiremnet Analysis	✓	✓							
System Design		✓	✓						
Frontend Development			✓	✓	✓				
Backend Development			✓	✓	✓				
AI Model Development				✓	✓	✓			
Federated Learning Module				✓	✓	✓			
Blockchain Integration					✓	✓	✓		
QR Verification Module					✓	✓			
Integration & Testing						✓	✓		
Performance Optimization							✓	✓	

Documentation							✓	✓	✓
Final Demonstration									✓

Comments

Industry, Innovation and Infrastructure

Approved by

Project guide	**Project Coordinator**	**HOD**
Ms. Neethu Roy	Ms. Meera V M	Dr.Manish T I
Assistant Professor	Assistant Professor	Professor
Dept of CSE	Dept of CSE	Dept of CSE
SSET	SSET	SSET